Module-1

Lecture-4: Deformation of Body

MECHANICS OF SOLIDS

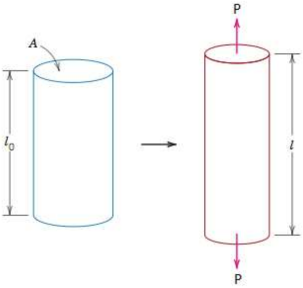
Content

- Deformation of body subjected to external force
- Deformation of body subjected due to self weight
- Principle of superposition
- Solved problems
- Assignment

Deformation of body subjected to external force

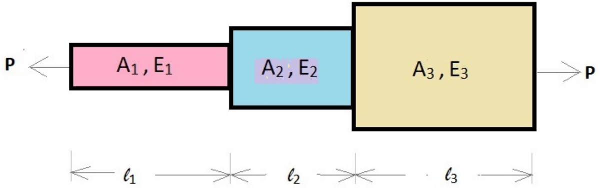
- Stress in the body is $\sigma = \frac{P}{A}$
- Strain is $\varepsilon = \frac{\delta l}{l}$
- Young's modulus $E = \frac{\sigma}{\varepsilon}$
- $E = \frac{\sigma}{\varepsilon} = \frac{\frac{P}{A}}{\frac{\delta l}{l}} = \frac{Pl}{A\delta l}$

$$\delta l = \frac{Pl}{AE}$$



Analysis of bars of varying sections

- Each section is subjected to same axial load P
- Stress and strain in each cross section is different
- Total change in length is the sum of change in length of individual section



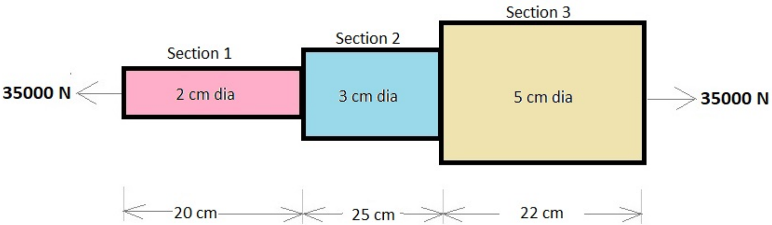
Total change in length of the bar

$$dL = P\left(\frac{l_1}{E_1A_1} + \frac{l_2}{E_2A_2} + \frac{l_3}{E_3A_3}\right)$$

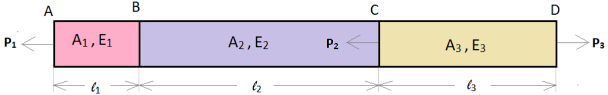
Workout

An axial pull of 35000 N is acting on a bar consisting of three lengths as shown in figure below. If the Young's modulus is 210 kN/mm^2, determine

1. Stress in each section
2. Total extension of the bar



- Sometimes body will be subjected to a number of forces
- FBD of individual sections are analysed

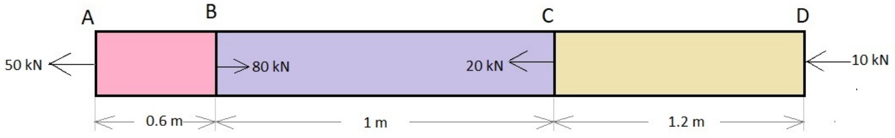


Principle of superposition

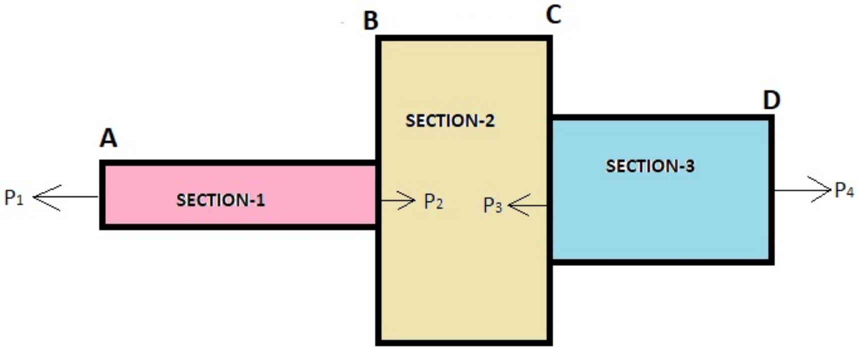
The resulting deformation of body = algebraic sum of deformation of individual sections

$$\delta l = \frac{Pl}{AE} = \frac{1}{AE}(P_1 l_1 + P_2 l_2 + P_3 l_3 + \ldots)$$

A brass bar, having cross sectional area of 1000 mm^2, is subjected to axial forces as in figure. Find the total elongation of the bar. Take E= 105 kN/mm^2



Calculate the force P_2 necessary for equilibrium, if $P_1 = 45kN, P_3 = 450kN$ and $P_4 = 130kN$.
Determine the total elongation of the member.
Given data:
$l_1 = 120cm, l_2 = 60cm, l_3 = 90cm, A_1 = 625mm^2, A_2 = 2500mm^2, A_3 = 1250mm^2, E = 210GPa$



Thank you!